

Course Code : ECT 451-2

GVHW/RW – 23 / 1025

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

MICROWAVE THEORY AND TECHNIQUES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Due credit will be given to neatness and adequate dimensions.
- (4) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1.
 - (a) Outline a short history of Microwave Engineering Development. 5(CO1)
 - (b) Translate the First Five IEEE Microwave Bands in terms of its wavelength. 5(CO1)

2.
 - (a) Summarize Planar type Transmission Lines. 6(CO2)
 - (b) In a unmatched transmission line, the total power incident on Port 1 is 5 W, whereas 2.2 W is reflected back. Power delivered to the load is 2 W at Port 2. Solve for (in dB) :
 - (i) Attenuation Loss.
 - (ii) Reflection Loss.
 - (iii) Insertion Loss. 4(CO2)

3.
 - (a) Recall application of Faraday Rotation in Ferrites using the working of Gyrator. 5(CO4)
 - (b) A signal of 32 W is fed into one of the collinear ports of a loss-less H-Plane Tee junction. Discover the powers in the remaining ports when others are terminated by means of matched load. 5(CO4)

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4. (a) A Circular MSA is to be designed at 2.45 GHz on a substrate having $\epsilon_r = 2.55$, $h = 1.66$ mm, $\tan\delta = 0.0012$. Predict the radius of a circular antenna and feed point location for $50\ \Omega$ impedance matching. 6(CO3)
- (b) Outline Periodic Structures in transmission lines. 4(CO3)
5. (a) Illustrate Antenna Gain measurement setup in anechoic chamber with block diagram. 6(CO5)
- (b) Two identical 30 dB directional coupler are used to sample incident and reflected power in a waveguide with $VSWR = 2$. The output of the coupler sampling incident power is 4.5 mW. Estimate the value of reflected power. 4(CO5)
6. (a) Distinguish between Continuous Wave Doppler and Pulsed RADAR. 5(CO1)
- (b) Estimate the range of a RADAR system which operates at 3 cm with a peak pulse power of 600 KW, if its antenna is 5 m^2 , minimum detectable signal is 10^{-13} W and the cross sectional area of the target is 20 m^2 . 5(CO1)

